

Gas Usage Prediction System using Mathematical Analysis

Dataset: sample_gas_usage.csv

Target column: gas_usage_m3

Linear regression equation: $y = -4.2354x + 407.8527$

Metric	Value
Total Gas Usage	13129.70
Average Consumption	364.71
Predicted Future Usage	326.62
Monthly Growth Rate	-3.22
Daily Forecast	326.62
Weekly Forecast	329.52
Monthly Forecast	340.64

Forecast generated using linear regression, moving average, exponential smoothing, and trend analysis.